PPM Dies Monitoring System - Quick Reference

System Purpose

Objective: Control system to protect die maintenance from exceeding stroke limits and reduce problematic parts due to die condition issues.

Key Formulas

Standard Stroke Priority

```
1. control_stroke (if set)
2. ppm_standard (if set)
3. tonnageStandard.standard_stroke (from machine)
4. Default: 10,000
```

Lot Calculation

```
Total Lots = ceil(Standard Stroke / Lot Size)
Example: ceil(10,000 / 2,500) = 4 lots
```

Alert Thresholds

```
Orange Threshold = Standard Stroke - Lot Size
Red Threshold    = Standard Stroke

Example with Standard=10,000, LotSize=2,500:
- Orange at ≥ 7,500 strokes
- Red at ≥ 10,000 strokes
```

Zone Coloring

Position	Zone	Color	Meaning
Last lot	Red	🔴	Critical - Stop Production
Second-to-last	Orange	🟠	Warning - Schedule PPM
Others	Green	🟢	Normal - OK to continue

Alert Recipients

Alert Type	Recipients
🟠 Orange	MGR/GM, MD
🔴 Red	MGR/GM, MD, MTN Dies

User Roles

Role	Code	Primary Function
Admin	`admin`	Full system access
Production Engineering	`pe`	Production planning
Maintenance Dies	`mtn_dies`	Die maintenance
Manager/GM	`mgr_gm`	Approval & monitoring
Managing Director	`md`	Executive oversight

Key Files

Purpose	Location
Die Model (Core Logic)	`app/Models/DieModel.php`
Alert Service	`app/Services/AlertService.php`
Notifications	`app/Notifications/CriticalDieAlert.php`
Die Controller	`app/Http/Controllers/DieController.php`
Production Controller	`app/Http/Controllers/ProductionLogController.php`
Lot Progress Component	`resources/js/Components/PPM/LotProgress.jsx`

Production Log Impact

On Create

```
die.accumulation_stroke += output_qty
```

On Update

```
difference = new_output - old_output
die.accumulation_stroke += difference
```

On Delete

```
die.accumulation_stroke -= output_qty
```

PPM Processing

1. **Schedule PPM** → Set planned date
2. **Start PPM** → Mark in-progress
3. **Complete PPM** →
    - Reset accumulation_stroke = 0
    - Clear ppm_alert_status = 'none'
    - Record in PPM History

Generated: January 28, 2026